DataSetu: India Public Data Analytics Platform

(Education Module – v1.0)

Database Systems Lab (BCSE302P)

PROJECT REPORT

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CONTENTS

Project Title

1. **Short summary of the project**
2. **Front end and Database details**

DataSetu: India Public Data Analytics Platform (Education Module)

1. Short Summary of the Project

Problem Statement

Publicly available Indian government education datasets (such as UDISE and UDISE+, published via the National Data & Analytics Platform - NDAP) are rich in information but exist as scattered, raw CSV files with inconsistent formats. There is no single, structured, analytics-ready platform that transforms this raw government data into meaningful business intelligence for researchers, journalists, policy analysts, and UPSC aspirants. DataSetu addresses this gap by building an end-to-end data pipeline that converts raw government education data into a structured, queryable, and visually explorable analytics platform.

Objective

The objective of this project is to design and implement a relational database system that consolidates state-level Indian school education data, enabling meaningful Key Performance Indicators (KPIs) such as school infrastructure adequacy, teacher qualification distribution, pupil-teacher ratios, and academic performance outcomes, to be derived through structured SQL queries and visualized through interactive dashboards.

Data Source and Scope

The primary data source for this project is NDAP (National Data & Analytics Platform, NITI Aayog), accessed via its public API. Version 1.0 of this project is deliberately scoped to 13 datasets sharing a common grain (Country + State + Year, with some datasets further segmented by Gender or Management Type), drawn from two UDISE collections:

- Unified District Information System for Education (UDISE) - Flash Statistics: covering school counts by management type, teacher distribution, pupil-teacher ratios, retention rates, and examination results.
- Unified District Information System for Education Plus (UDISE+): covering school infrastructure - internet facility, computer facility, drinking water, and toilet facility (boys and girls), segmented by school management type.

Datasets that did not share this common state-level grain (such as District-level National Achievement Survey data, or Higher Education datasets from AISHE and NIRF describing a different entity altogether) were deliberately excluded from Version 1.0 and documented as scope for future versions, to ensure all included data can be reliably joined into a single, consistent schema.

Proposed Technology Stack

- Data Source: NDAP public API (JSON/CSV)
- Data Cleaning & ETL: Python (pandas, SQLAlchemy)
- Database: PostgreSQL
- Business Intelligence & Visualization: Microsoft Power BI
- Presentation Layer: Web-based dashboard (Power BI embedded)

Expected Outcome

The final deliverable will be a working PostgreSQL database populated with cleaned, structured education data, exposed through SQL views answering specific business questions

(e.g., infrastructure disparity across school management types, correlation between teacher qualification and examination outcomes), and visualized through an interactive Power BI dashboard, ultimately accessible via a public-facing website.

2. Front End and Database Details

Database Design Approach

The database is being designed using a dimensional modeling approach (star schema), suited for analytical querying and Power BI integration. The design separates descriptive, slowly-changing attributes (dimensions) from measurable, aggregatable data (facts):

Dimension Tables (proposed)

- Dim_State — State/UT name, region
- Dim_Year — Academic/financial year
- Dim_Gender — Male, Female, Total
- Dim_Management — Government, Government Aided, Private Unaided, Other Management types

Fact Tables (proposed)

- Fact_Schools — number of schools by state, year, and management type
- Fact_Teachers — teacher distribution by management type and academic qualification
- Fact_Infrastructure — school facility availability (internet, computer, drinking water, toilets) by management type
- Fact_ExamResults — Grade X and Grade XII examination appeared/pass percentages by management type
- Fact_Retention — retention rate by educational stage and gender

Data Cleaning and Loading Process

Raw CSV files obtained via the NDAP API are cleaned using Python (pandas) to resolve inconsistencies identified during manual data review, including non-uniform column naming across datasets (e.g., 'State' vs. 'State/Ut Name'), character encoding errors in column headers, and mismatched unit-of-measurement labels. Cleaned data is then loaded into PostgreSQL tables using SQLAlchemy, following the dimension-fact schema described above.

Query and Analytics Layer

SQL Views will be created on top of the fact and dimension tables to pre-compute answers to specific business questions, such as comparative infrastructure availability across government versus private schools, and the relationship between teacher qualification levels and examination pass percentages across states.

Front-End / Presentation Layer

Microsoft Power BI will connect directly to the PostgreSQL database (via the SQL Views layer, not raw tables) to build interactive dashboards featuring state-wise filters, year-over-year trend analysis, and comparative KPIs across school management types. These dashboards will subsequently be embedded into a public-facing website, making the analytics accessible to the target audience of researchers, journalists, policy analysts, and UPSC aspirants.

Current Status

At the current stage, all 13 datasets for Version 1.0 have been identified, downloaded, and are undergoing manual review to document column-level meaning and data quality issues. This review directly informs the finalization of the database schema, which is the immediate next step prior to PostgreSQL implementation.